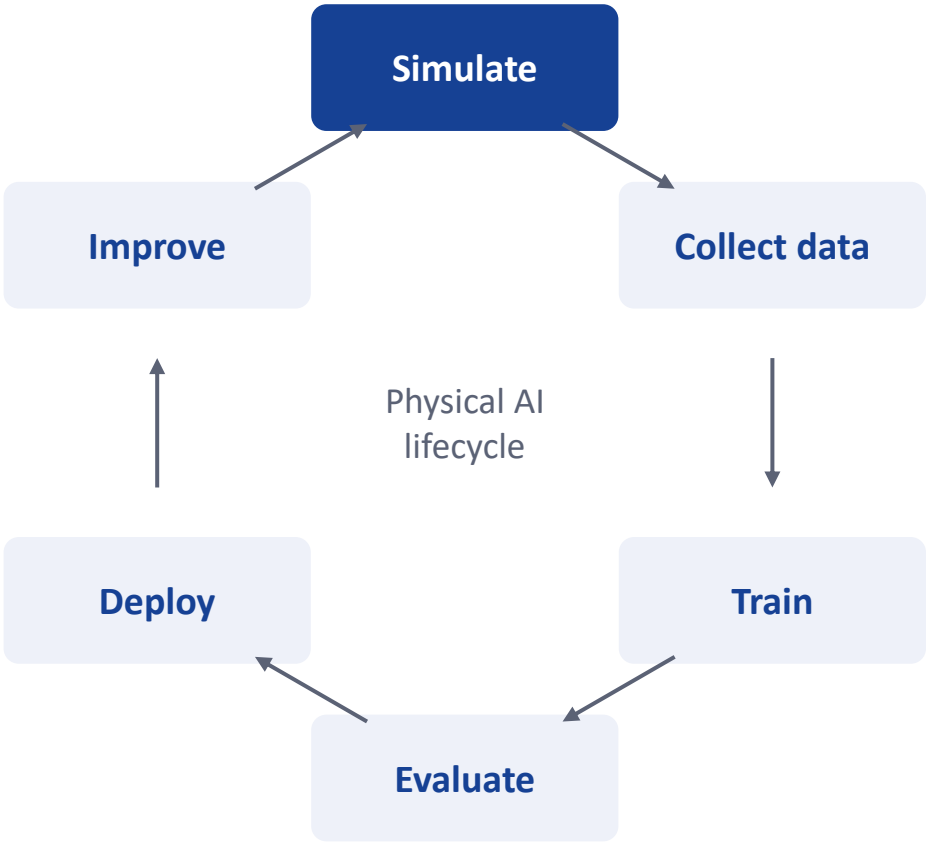
NANYANG
THE INNOVATIVE POLYTECHNIC

Physical AI: How a Robot Learns

Data Acquisition & Policy Learning · Lesson 1 (Lecture)



Our course map: one cycle, walked three times



- **A** Foundations L1–L2
- **B** Collecting robot data L3–L4
- **C** RL in simulation L5–L6
- **D** RL: sim → real arm L7–L8
- **E** Imitation learning L9–L10
- **F** Evaluate & deploy IL L11–L12
- **G** Vision-language-action L13–L14

Quick poll

What do you think is hard about getting a robot to do something useful?

Answer anonymously in the poll. One or two words is fine.

- **seeing?**
- **grasping?**
- **data?**
- **safety?**
- **cost?**

Learning outcomes

- By the end of this lesson you should be able to:
  - explain how physical AI differs from AI that runs on a screen
  - describe the six stages of the physical AI development lifecycle
  - name the observation and action of the SO-101 at each control step
  - compare reinforcement learning, imitation learning and VLA by where their training data comes from

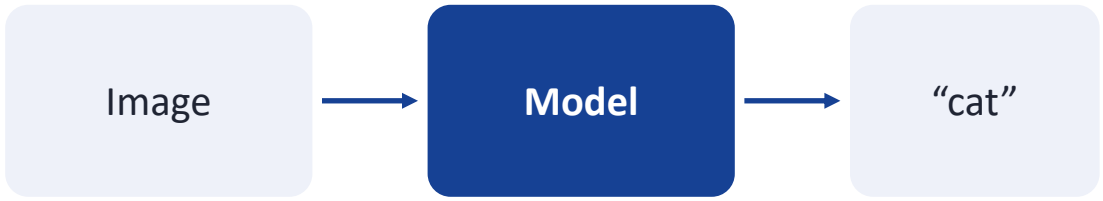
Part 1

From screen AI to physical AI

Why a robot is not just a classifier with a body

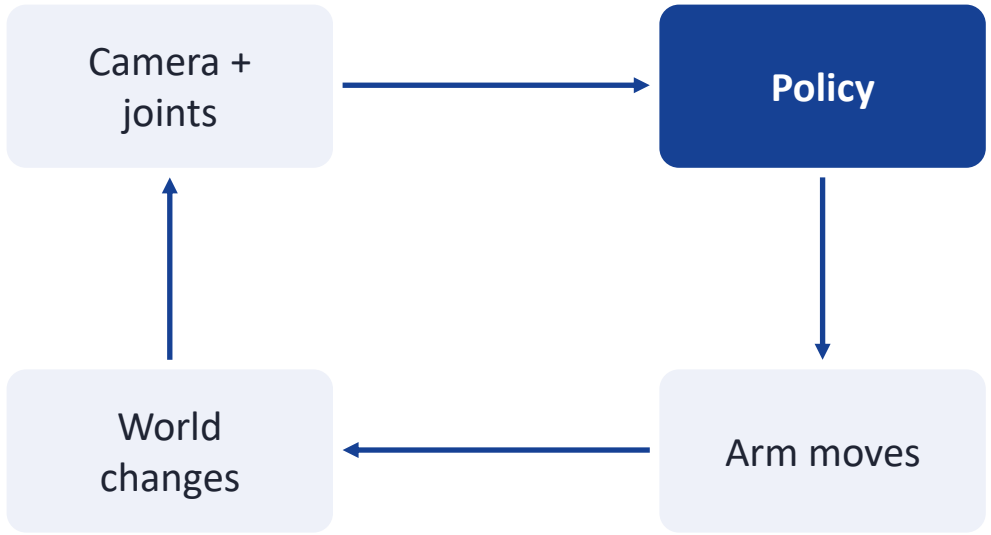
Screen AI answers once; a robot decides 30 times a second

Screen AI (e.g. image classifier)



One input, one answer.
The answer does not change the next image.
A mistake is just a wrong label.

Physical AI (robot policy)



Every action changes the next input.
A mistake moves the robot somewhere new.

The control loop: perceive → decide → act, every 33 ms



…and repeat. At 30 Hz there are 30 decisions per second: one every 33 milliseconds.

The policy is the only part we train. Everything else (the sensors and motors) stays the same for every method in this course.

Why physical AI is hard

Errors compound

2 mm off at the start of a grasp can become 2 cm off at the end. Each mistake feeds the next decision.

Data is not independent

Each frame follows from the last. Shuffling frames like images hides how the robot actually behaves.

Mistakes cost money

A bad action can hit the table, drop the object or strain a motor. Testing is never free.

Data is slow

One real demonstration takes about a minute. An image dataset can be scraped; robot data must be performed.

Simulation makes trials cheap, at a price

What simulation gives us

- Thousands of robots trying at once on one GPU
- Perfect labels for free (exact positions, contacts)
- No broken hardware, no safety risk
- Faster than real time

What it costs: the reality gap

Dynamics
masses, friction, how the motors respond

Appearance
colours, textures, lighting

Sensing
noise, offsets, delays

Anything the simulator leaves out, the policy never learns about.

Part 2

Robot basics with the SO-101

Joints, spaces, servos and one control step

Meet the SO-101

PHOTO / VIDEO
Labelled photo of your lab's SO-101 (leader and follower)

1. **shoulder_pan** rotates the whole arm left/right
2. **shoulder_lift** raises/lowers the upper arm
3. **elbow_flex** bends the elbow
4. **wrist_flex** tilts the wrist up/down
5. **wrist_roll** rotates the wrist
6. **gripper** opens and closes the jaws

Open-source design by TheRobotStudio with Hugging Face

Joints, degrees of freedom and workspace

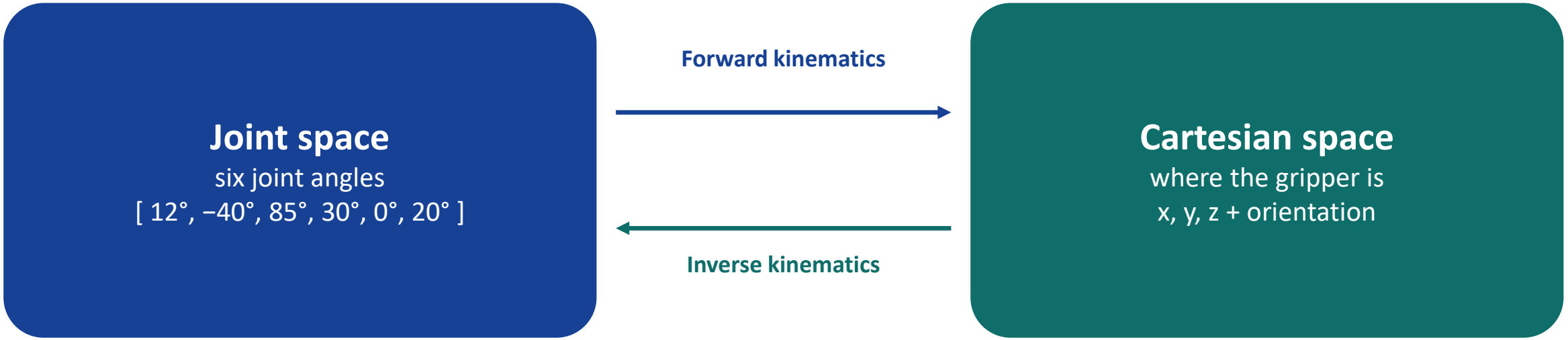
Joint
A place where two parts of the arm connect and can move.

Degree of freedom (DOF)
An independent way the robot can move. The SO-101 has 5 arm DOF plus a gripper.

Workspace
All the places the gripper can reach. Outside it, no policy can succeed.

PHOTO / VIDEO
Short clip: moving one joint at a time

Two ways to describe the same pose



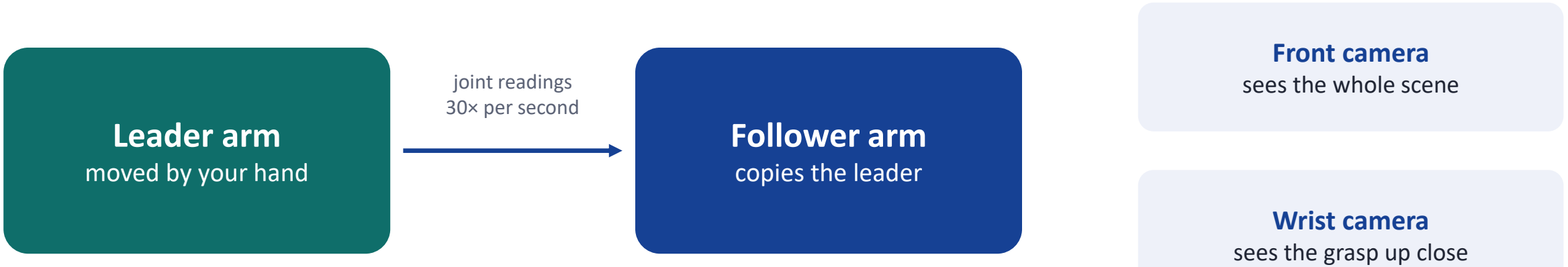
Forward: “Given how my arm is bent, where is my hand?” (one answer)
Inverse: “How should I bend my arm to reach the cup?” (maybe many answers, maybe none)

Our servos take positions, not forces

We CAN command	We CANNOT command
A target position for each joint (the servo's own controller drives towards it)	A force or torque “Grip harder” or “push gently”

Six Feetech STS3215 serial-bus servos. Grip force comes only indirectly, by telling the gripper to close further than the object allows. This limit returns in Lessons 7 and 9.

Leader, follower and cameras



Recorded every frame: leader position (the action), follower position (the state), both camera images.
This is how most of the data in this course will be collected.

Worked example: one control step

Observation (input)

6 joint positions
[2.1, −38.5, 81.0, 27.4, −1.2, 18.0]
+ front image + wrist image

policy →

Action (output)

6 target joint positions
[2.4, −37.9, 80.2, 27.9, −1.2, 12.5]

180
joint numbers per second

60
images per second

600
frames in a 20 s demo

1,200
images in a 20 s demo

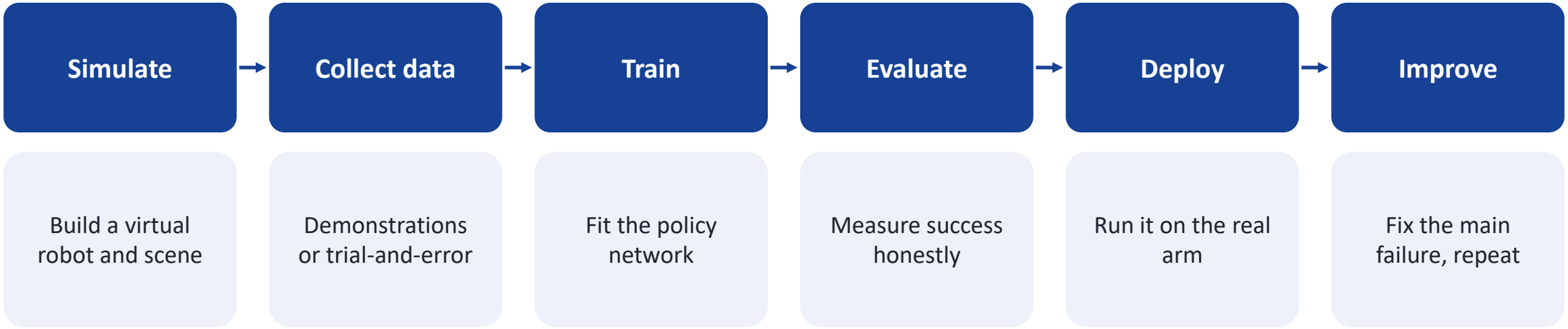
Example values are illustrative; replace with numbers captured from your own arm.

Part 3

The lifecycle and three ways to train an arm

Where does the training data come from?

The physical AI development lifecycle



Simulation helps most at the start. Evaluation and improvement are where most real projects spend their time.

Three ways to train an arm

	Reinforcement learning	Imitation learning	Vision-language-action
Where data comes from	Robot tries, gets a score	Human demonstrates	Demos + a sentence
What you provide	You write the reward	You record the dataset	Fine-tune a big pretrained model
Where it trains	Mostly in simulation	Real arm, no reward	One model, many tasks
Typical failure	Fails by reward hacking	Fails off its training data	Can pick the wrong task

The key question: does the data come from the robot trying, or from a person showing?

Course roadmap: 14 weeks

Foundations		Data		RL in sim		RL → arm		Imitation		Evaluate		VLA	
Wk 1 **L1** Lec	Wk 2 **L2** Lab	Wk 3 **L3** Lec	Wk 4 **L4** Lab	Wk 5 **L5** Lec	Wk 6 **L6** Lab	Wk 7 **L7** Lec	Wk 8 **L8** Lab	Wk 9 **L9** Lec	Wk 10 **L10** Lab	Wk 11 **L11** Lec	Wk 12 **L12** Lab	Wk 13 **L13** Lec	Wk 14 **L14** Lab

Test 1
after Week 8 · Lessons 1–8

Test 2
after Week 14 · Lessons 9–14

Assignment
pick-and-place project · demo after Week 14

Lecture weeks explain the idea; lab weeks make it run in Isaac Sim and on the SO-101. Each lesson starts with a short check on the last one.

Activity: Spot the stage

In pairs, label each card with a lifecycle stage and the method it belongs to (RL, IL or VLA).

“A student hand-guides the leader arm fifty times.”
Answer: Collect data · IL

“4,096 simulated arms try to reach a target.”
Answer: Train · RL

“The policy succeeds 14 times out of 20 on the real arm.”
Answer: Evaluate · any

“Each demo is saved with ‘pick the blue cube’.”
Answer: Collect data · VLA

10 minutes · 8 cards per pair · answers shown on screen afterwards

Exit ticket

- Answer on paper or in the form before you leave:
  - What does the SO-101 policy receive as its observation each step?
  - What does it output as its action?
  - Put the six lifecycle stages in order.
  - Where does reinforcement learning get its training data?
  - Where does imitation learning get its training data?

Next week: L2 lab · Meet the tools

Sim station · Isaac Sim

Tour the simulator
Load the SO-101 and move each joint
Watch a trained Reach policy

Arm station · SO-101

Calibrate leader and follower
Teleoperate with live camera view
Fill in the joint table

Safety first: clear workspace, hands out of the follower's path, cut the power (not just USB) to stop, never leave a powered arm alone.

References and further reading

- Capuano et al. (2025). Robot Learning: A Tutorial. https://arxiv.org/abs/2510.12403
- Hugging Face LeRobot documentation: SO-101. https://huggingface.co/docs/lerobot/so101
- NVIDIA. Train an SO-101 Robot From Sim-to-Real With NVIDIA Isaac. https://docs.nvidia.com/learning/physical-ai/sim-to-real-so-101/latest/index.html
- Zhao et al. (2023). Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware (ACT). https://arxiv.org/abs/2304.13705
- Rudin et al. (2021). Learning to Walk in Minutes Using Massively Parallel Deep RL. https://arxiv.org/abs/2109.11978
- Lynch & Park. Modern Robotics (free book and videos). https://hades.mech.northwestern.edu/index.php/Modern_Robotics

Advanced topic

From one arm to a humanoid

Today's four ideas, applied to a robot shaped like a person

Extension material: the control loop, robot basics, the lifecycle and the three training methods, on a full humanoid.

Why build a robot shaped like a person?

1

Our world is built for human bodies

Stairs, door handles, shelves and tools all assume a human shape. A humanoid can use them without rebuilding the workplace.

2

People can show it what to do

A person's movements map naturally onto a human-shaped robot, so it is easier to demonstrate a task to it.

3

One body, many jobs

Instead of one special machine per task, one general robot that learns new tasks from data.

The catch

Everything on slide 8 gets harder: more joints, more ways for errors to compound, more expensive mistakes, and now the robot can fall over.

Still a control loop, now with layers inside it

Decide: the learned policy
looks at cameras and joints · chooses what the hands and body should do · about 30 times a second

hand targets, plus a walking command

Balance controller
keeps the robot upright while it moves · about 50 times a second

joint targets for the whole body

Motor controllers
hold each joint at its target · up to about 1,000 times a second

Compare with the SO-101 (slide 7)

One loop: perceive → decide → act, 30 times a second. The servos do the rest, and the arm is bolted to the table.

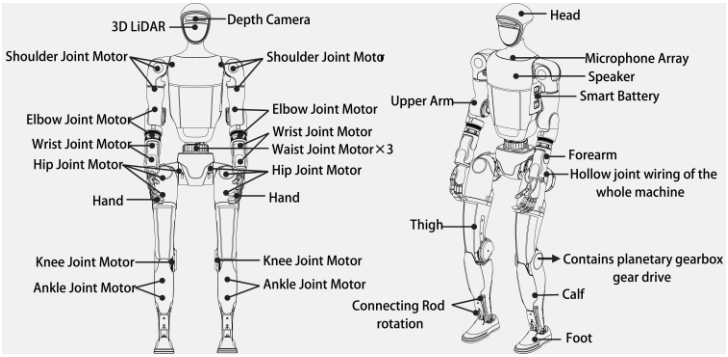
Why the extra layer?

A robot on two feet is always slightly falling. Balance needs fast, constant corrections, so it gets its own controller, usually trained with RL in simulation.

Slide 6 still holds: every action changes what the robot sees and feels next.

Meet a humanoid: joints and degrees of freedom





Head
cameras; 0–3 DOF neck

Each arm
5–7 DOF

Each hand
up to 7 DOF (fingers)

Waist
1–3 DOF

Each leg
6 DOF: hip ×3, knee, ankle ×2

6
SO-101
5 arm DOF + gripper

23
Unitree G1 (basic)
6 per leg, 1 waist, 5 per arm

43
Unitree G1 EDU
with extra waist, wrist and finger DOF

50
Boston Dynamics Atlas
electric humanoid, 2025

Workspace: legs move the whole workspace, so the robot can walk to the object.

Worked example: one control step on a humanoid

Observation (input)

Joint positions and speeds: about 30 joints

Body tilt and spin: from an IMU, the robot's sense of balance

Camera images: a stereo pair from the head, sometimes wrist cameras

policy →

Action (output)

Arm and hand targets: 28 numbers on a Unitree H1 (7 per arm, 6 per hand, 2 neck)

A walking command: forward speed, sideways speed and turn rate, sent to the balance controller

6 → 28
action numbers per step

180 → 840
action numbers per second

+1 sense
an IMU for balance

+1 layer
a balance controller

SO-101 → humanoid. The 28 action numbers are from the Unitree H1 in Open-TeleVision (2024); other robots differ.

New senses and different muscles

New senses	Different muscles
IMU measures tilt and spin, like your inner ear	**Position + stiffness** motors take a target position and how stiffly to hold it, so a joint can be firm or soft
Depth camera and 3D LiDAR measure how far away things are (both on the Unitree G1)	**Much stronger** a Unitree G1 knee motor gives up to 90–120 N·m of torque
Touch sensors in the hands optional on some dexterous hands, to feel a grasp	**Some take torque directly** “push this hard”: the command your SO-101 servos cannot take

Compare slide 14: the SO-101's servos take positions only. A humanoid needs gentler, stronger, more controllable joints because it touches the ground, objects and sometimes people.

The lifecycle, humanoid edition

Simulate	Collect data	Train	Evaluate	Deploy	Improve
Thousands of virtual humanoids learn to walk at once on one GPU	A person in a VR headset sees through the robot's eyes and moves its arms	RL in simulation for balance; imitation learning from demos for hand skills	In simulation first, then on a safety harness, counting successes	Policy on a GPU in the robot, balance controller underneath	Atlas dropped parts, so engineers recorded recovery demos and retrained

The same six stages as slide 18. Safety makes Evaluate and Deploy much slower on a humanoid.

Three ways to train a humanoid

	Reinforcement learning	Imitation learning	Vision-language-action
What it learns	walking and balance	hand skills: grasp, fold, sort	many tasks from spoken instructions
Where data comes from	millions of simulated tries and falls	a person demonstrating in VR	huge sets of demos, videos and sentences
Real example	HumanPlus: a body controller learned from 40 h of human motion	Open-TeleVision: sorting cans after VR demos	Figure Helix (~500 h of demos), NVIDIA GR00T N1

Real humanoids combine all three: RL for the legs, imitation for the hands, VLA to follow instructions.

A real example: HumanPlus combines RL and imitation

1

Learn to move (RL, sim)

The robot learns to copy 40 hours of recorded human motion in simulation, then moves to the real robot.

2

Shadow a person

One webcam watches the operator; software estimates their pose; the robot copies it live.

3

Record demos

30–40 demonstrations per task, seen through the robot's two head cameras.

4

Train by imitation

A policy based on ACT (Lesson 9) learns to do the task alone.

Results on a Unitree H1 (33 joints, including hands)

Fold clothes: 100% (40 demos)

Type "AI" on a keyboard: 80% (30 demos)

Put on a shoe and walk: 60% (40 demos)

Spot the methods

Step 1 is reinforcement learning: the robot tries. Steps 2–4 are imitation learning: a person shows. Each task needed about 40 demos, close to the 50 you will record.

Industry scale: the same ideas with far more data

Boston Dynamics Atlas

450M

parameter policy

Operators in VR with trackers on chest and feet demonstrate whole-body tasks.

One policy reads head cameras, joints and a sentence describing the task.

Figure Helix

≈500 h

of demonstrations

A ‘slow thinker’ understands the scene and instruction about 8 times a second.

A ‘fast doer’ moves 35 joints, including fingers, 200 times a second.

NVIDIA GR00T N1

750K

simulated demos in 11 hours

An open humanoid model trained on human videos, simulation and real demos.

You fine-tune it with your own small dataset.

Bigger models, much more data, plus language: Lessons 13–14 (VLA) are where this course meets these systems.

What you would need: hardware and software

Tier 1 · Simulation only
start here, with next week's tools

Hardware: an NVIDIA RTX PC like our Isaac Sim machines.

Software: Isaac Lab, which includes humanoid robots such as the Unitree G1 and Fourier GR-1, plus humanoid imitation-learning examples.

Tier 2 · Research lab humanoid
a realistic polytechnic or university setup

Hardware: Unitree G1 (basic model from US$13.5K; the EDU version costs more), a VR headset such as Apple Vision Pro or Meta Quest 3, one strong GPU.

Software: LeRobot, the same library as the SO-101, now supports the Unitree G1.

Tier 3 · Industrial
company scale

Hardware: full-size humanoids (Atlas, Figure), teams of operators, GPUs inside the robot, GPU clusters for training.

Software: in-house foundation models, or open ones such as GR00T N1, fine-tuned.

The common thread: Isaac Sim, LeRobot and the same lifecycle you start using in Lesson 2.

Discussion: from the SO-101 to a humanoid

1. **Outcome 1 ·** Why does a humanoid need a balance layer in its control loop when the SO-101 does not?
2. **Outcome 3 ·** A humanoid picks up a cup while standing. Name its observation and its action at one control step.
3. **Outcome 2 ·** Which lifecycle stage would take longest on a humanoid, and why?
4. **Outcome 4 · A humanoid must carry a box across a room when asked. Which part would you train with RL, which with imitation learning, and where does VLA help?**

Pairs, 10 minutes. Be ready to share one answer with the class.

References: advanced topic (humanoids)

Cheng, X. et al. (2024). Open-TeleVision: teleoperation with immersive active visual feedback. https://arxiv.org/abs/2407.01512

Fu, Z. et al. (2024). HumanPlus: humanoid shadowing and imitation from humans. https://arxiv.org/abs/2406.10454

Boston Dynamics (2025). Large behavior models and Atlas find new footing. https://bostondynamics.com/blog/large-behavior-models-atlas-find-new-footing/

Figure AI (2025). Helix: a vision-language-action model for generalist humanoid control. https://www.figure.ai/news/helix

NVIDIA et al. (2025). GR00T N1: an open foundation model for generalist humanoid robots. https://arxiv.org/abs/2503.14734

Unitree G1 humanoid: specifications. https://www.unitree.com/g1

LeRobot documentation: Unitree G1. https://huggingface.co/docs/lerobot/unitree_g1

Isaac Lab documentation: data generation and imitation learning for humanoids. https://isaac-sim.github.io/IsaacLab/main/source/features/imitation-learning/humanoids_imitation.html